PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 1 · How this class works, and a statistics review

TUESDAY AUG 25 AND THURSDAY AUG 27, 2026

START OF CLASS

EVERY TUESDAY STARTS THE SAME WAY: PULL THE NEW COURSE MATERIALS

1. Open a terminal and go to the course-materials folder.

   cd spax402, then cd spax402-course-materials

2. Pull the week's new files.

   git pull

3. That is where every deck and data file arrives.

   Your own repo holds your work; this one holds the handouts. Today you will clone it during setup, so the ritual starts for real next Tuesday.

INTRODUCE YOURSELF.

Name, major, and year.

Why you registered for this class.

The sport you follow most closely, and the job you would take tomorrow if somebody offered it.

01

HOW THIS CLASS WORKS

MY GOAL: EQUIP YOU WITH THE INTUITION, CRITICAL THINKING, TECHNICAL, AND COMMUNICATION SKILLS NEEDED TO USE ANALYTICS TO ADD VALUE TO AN ORGANIZATION.

EVERY UNIT HAS FOUR PARTS

1. **Tuesday's lecture.**

   The unit's concepts, in class.

2. **Thursday, in class, by hand, no agentic AI.**

   We will use Excel to do the calculations and build an understanding of our approaches. Without a mental model of what we're doing and why, you will misapply models to the wrong contexts and communicate their outputs incorrectly. You may still use an LLM to assist with troubleshooting formulas and understanding our approach.

3. **The take-home Case Study. On your own, with Claude Code.**

   You will practice using agentic coding tools to quickly run similar analysis more efficiently. When complete, you will communicate your findings in your own words, along with submitting the code.

4. **Tuesday's quiz, closed-book.**

   A short quiz at the start of Tuesday's class to confirm that you understand the concepts.

ASSIGNMENT ASSISTANCE RESTRICTIONS

1 Thursday, in class, by hand, no agentic AI.

You may use an LLM to assist with troubleshooting formulas and understanding our approach. The hand-build may be finished at home that evening; it is submitted by Monday at 11:59pm, along with the AI-assisted assignment and brief.

2 The take-home Case Study. On your own, with Claude Code.

Not allowed: submitting analysis created or edited by another human. You may discuss assignments with classmates, and use books, instructors, and LLMs. Not allowed: submitting briefs written by AI or another human. The BRIEF.md submissions must be in your own words; copying and pasting from an LLM and changing some of the verbiage is also in violation. You need to be able to sit in a room with your boss, a coach, or a GM and explain your findings.

3 Tuesday’s quiz, closed-book.

Quizzes and both exams are closed-book and proctored. If you can only explain a concept by consulting an outside resource, you don’t understand the concept yet.

GRADING

COMPONENT	WEIGHT	NOTES
Quizzes	25%	Weekly, Tuesday, closed-book. Two lowest dropped.
Case studies	25%	Equal halves: the analysis and the brief. Audit and quiz artifacts required.
Exams	25%	Exam 1 written (Oct 8). Exam 2 is a live audit (Nov 19).
Final project	25%	Proposal Week 11, presented Dec 8. No final exam.

GRADING

FOUR COMPONENTS OF EACH CASE STUDY

1. **The analysis.**

   The spreadsheet, code, chart, table. Whatever you built, including Thursday's hand-built file.

2. **The quiz transcript.**

   /quiz-me quizzes you on that week's concepts and writes the whole exchange into your repo. This will prepare you for the in-class quiz on Tuesday.

3. **The audit record.**

   /audit reviews your analysis like a skeptical senior analyst and writes down each finding. You respond to every one – fix it, or defend why it stands – in the same file.

4. **The brief.**

   BRIEF.md, half a page to the week's audience, typed by you.

HOW A CASE STUDY IS GRADED

1. **Half: the analysis.**

   The spreadsheet, code, chart, table. Whatever you built, including Thursday’s committed hand-built file.

2. **Half: BRIEF.md, typed by you.**

   Half a page to the week’s named audience, usually a coach. Graded on decision quality and on honesty about what you do not know. You may ask Claude to red-team the draft; you may not ask it to write the draft.

3. **Required but not graded: the audit record and quiz transcript.**

   These steps will help catch an issue with your analysis before I do. They’ll also prepare you for your quiz.

CALENDAR

THE FIRST HALF OF THE SEMESTER

WK	TUE / THU	CONTENT
1	Aug 25 / 27	Onboarding and statistics review
2	Sep 1 / 3	Classical probability and inference
3	Sep 8 / 10	Bayesian statistics
4	Sep 15 / 17	Descriptive and probabilistic statistics
5	Sep 22 / 24	Introduction to regression
6	Sep 29 / Oct 1	Regression continued
7	Oct 6 / 8	Review Tuesday. Exam 1 Thursday.

First case study due Mon Aug 31. Labor Day pushes Week 2's to Tue Sep 8. Sep 8 is the last add or drop day.

CALENDAR

THE SECOND HALF OF THE SEMESTER

WK	TUE / THU	CONTENT
8	Oct 13 / 15	Data collection and responsible automation
9	Oct 20 / 22	Storytelling, visualization, skills and hooks
10	Oct 27 / 29	Logical fallacies and AI shortcomings
11	Thu Nov 5 only	Monte Carlo. Final project proposal due.
12	Nov 10 / 12	Machine learning
13	Nov 17 / 19	Practical exam: three flawed analyses, audited live
14	Dec 1 / 3	Final project work sessions
15	Tue Dec 8	Final presentations

Election Day suspends classes Tue Nov 3. No class Nov 23 to 27 for Thanksgiving. Final deliverable due Mon Dec 14.

02

OUR TECH STACK

Today is setup. Thursday you use all of it on real data.

THREE THINGS BEFORE YOU LEAVE TODAY

1 **Claude Pro, installed and signed in.**

Roughly twenty dollars a month for four months. If that is a real obstacle, email me privately and it gets handled. Nobody is dropping this course over a subscription.

2 **Claude Code and Git on your machine.**

Half an hour with the setup page, longer if you have never used a terminal. Do it before class if you can. Stragglers get fixed Thursday, not left behind.

3 **Your repository, created from the course template on GitHub.**

This is where every assignment lives for the rest of the semester. Your copy is yours; I can see it, your classmates cannot.

THE SAME FIVE MOVES EVERY WEEK

1. **Hand-build the method.**
   Thursday, in class, no agentic AI. One team, ten trials, two hypotheses.

2. **Scale it with the agent.**
   Your artifact is the specification: the target, the assumptions, the transformations, the evaluation, the outputs you want. Claude writes the Python. You read the outputs, not the code.

3. **Verify adversarially.**
   Run /audit. Check the agent's answer against your Thursday hand-build. If they disagree, one of you is wrong.

4. **Brief the week's audience.**
   BRIEF.md, half a page, typed by you. What you found, what you would do, and what would change your mind.

5. **When you are done, run /submit.**
   It checks the whole week – hand-build, analysis, audit, brief, quiz – tells you what is missing, and pushes. You can always submit; anything missing at the deadline costs you on the analysis half of the grade. Repo link in Canvas by Monday 11:59pm.

THE FOUR COMMANDS YOU WILL USE MOST

/teach

Explains a concept using the week's data.

/audit

Has the agent audit your own work like a skeptical senior analyst, re-deriving every key number from the raw data. Run this before you push your assignments.

/quiz-me

Quizzes you on that week's concepts before you submit your assignment. This will prepare you for the in-class quiz on Tuesday.

/submit

Submits your work. Before doing so, it checks every required file and tells you if anything's missing.

SETUP TIME.

1. Install Git and Python.
2. Install the Claude Desktop app. Sign in.
3. GitHub account. Create your repo from the template. Add jackdav1.
4. Install the Python packages.
5. First commit, pushed.

Full instructions: SPAX-402-Setup on Canvas. Flag me when something breaks — we continue once everyone is running.

03

THE SHAPE OF THE DATA

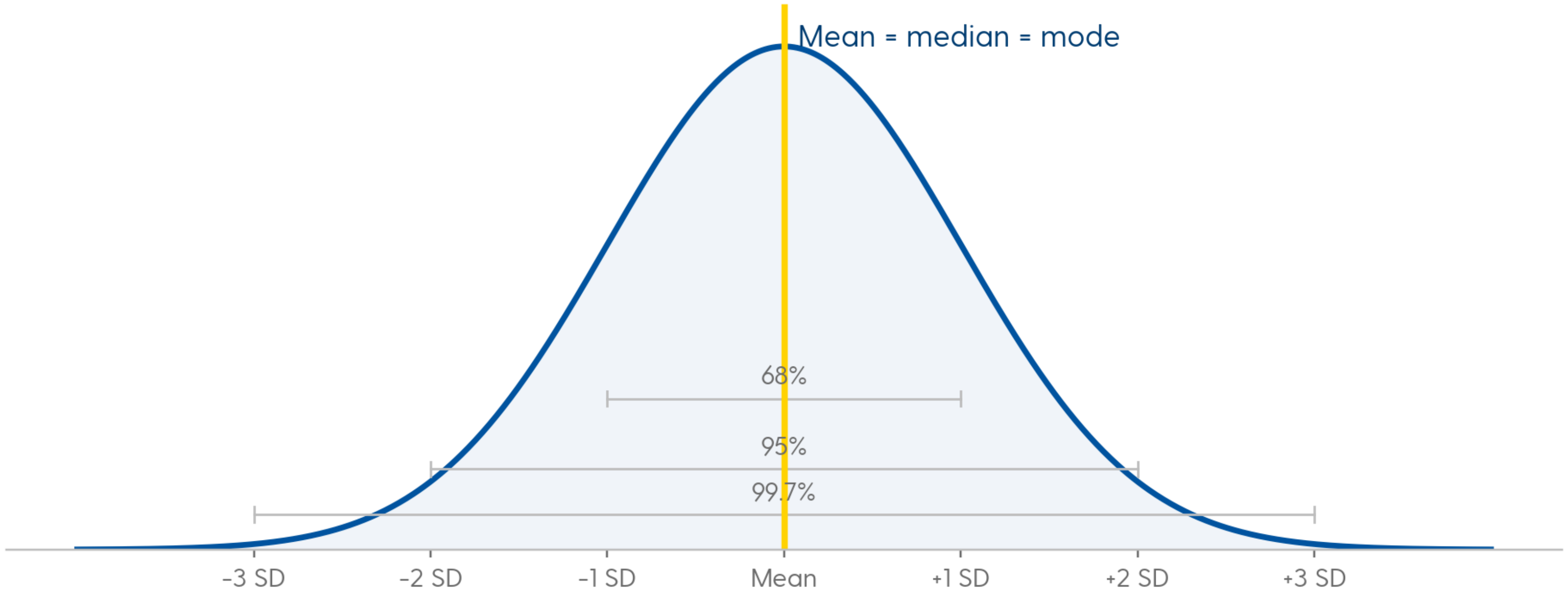
Mean, median, mode, and the reason the first question about any statistic is what its distribution looks like.

MEAN, MEDIAN, MODE

The average, the middle value, and the most common value. In a symmetric distribution they land on the same number; how far apart they sit is a measure of how skewed the data is.

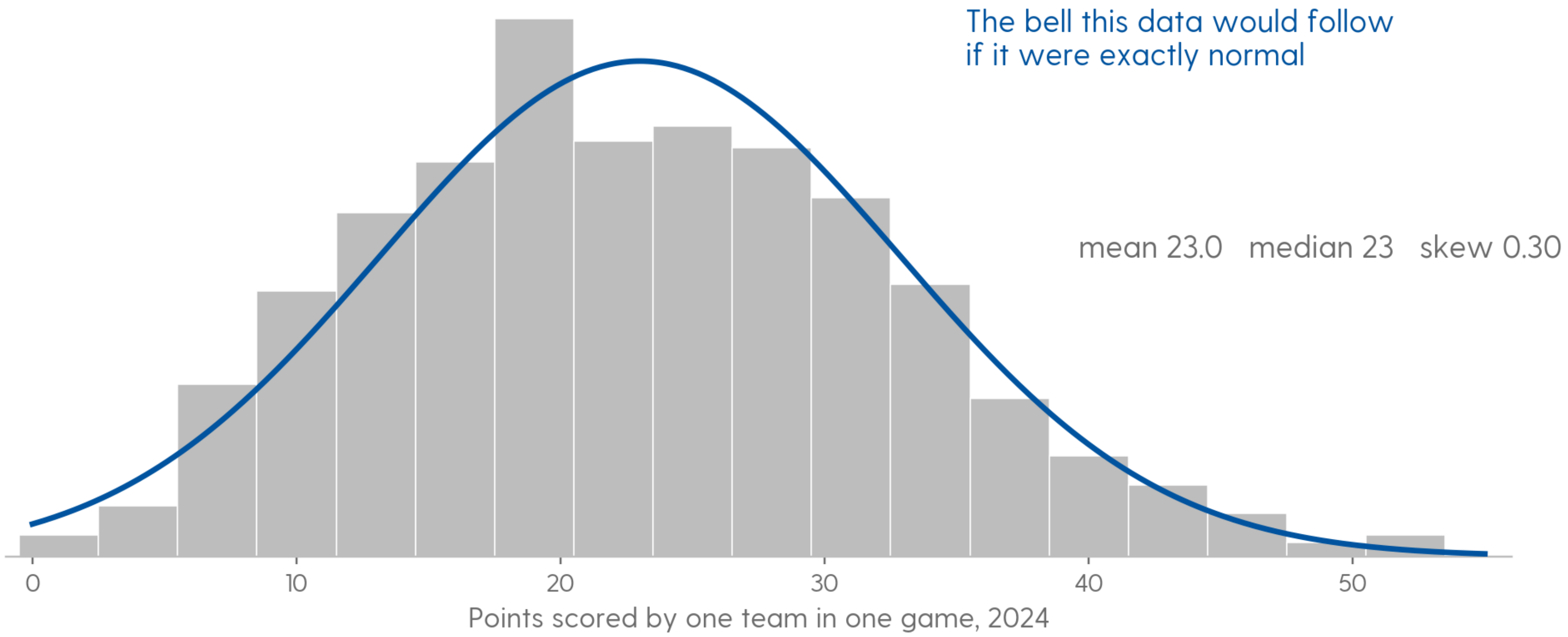
Mean = $\Sigma x / n$. Median = the 50th percentile. Mode = the most frequent value, and the only one of the three that also works on categories.

WHERE THE THREE-SIGMA RULES COME FROM



Symmetric, one peak, mean equals median equals mode. Roughly 68, 95, and 99.7 percent of values inside one, two, and three standard deviations. Every one of those numbers is a property of this exact shape.

POINTS PER TEAM-GAME IS CLOSE TO NORMAL



Mean 23.01, median 23, skew 0.30 across 570 team-games. Close enough to normal that the sigma rules roughly hold. Most sport statistics are not.

nflverse play-by-play, 2024 regular season and playoffs

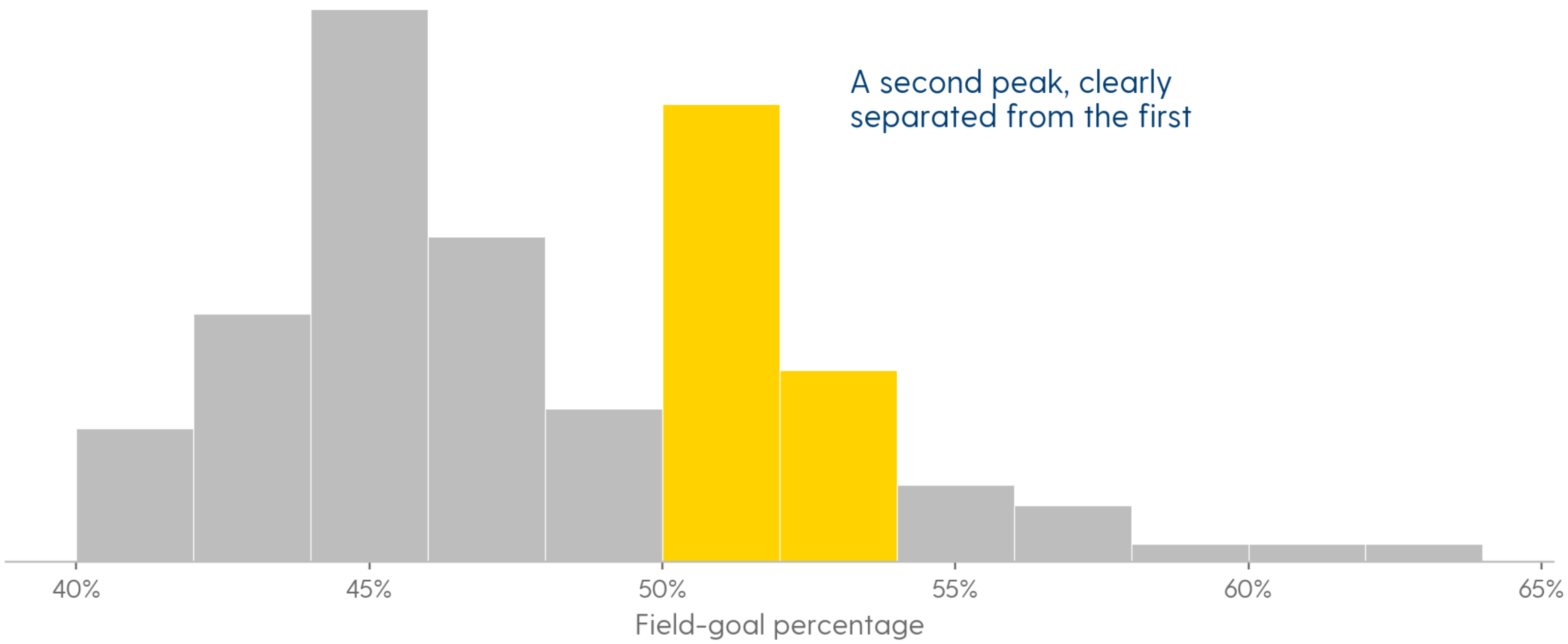
? RUSHING YARDS PER CARRY: DO YOU EXPECT THE MEAN TO BE HIGHER OR LOWER THAN THE MEDIAN?

WHICH OF THESE STATISTICS DO YOU EXPECT TO BE SKEWED, IN WHICH DIRECTION, AND WHY?

Home runs. Free-throw percentage. Golf scores. Time to recover from an injury. Height. Salary.

For each one: is there a floor, is there a ceiling, and what would it take for a single athlete to sit far out on the tail?

TWO PEAKS USUALLY MEAN YOU HAVE MIXED TWO DIFFERENT THINGS TOGETHER



NBA field-goal percentage, every qualifying player. One peak in the mid-40s, a second at 50 to 52. A distribution with two modes is bimodal, and the second mode is a question that needs an answer.

Severini (2020), Analytic Methods in Sports

WHAT DO YOU THINK THAT SECOND PEAK OF PLAYERS ON THE RIGHT IS?

A BIMODAL DISTRIBUTION MEANS YOUR DATA LIKELY HAS MULTIPLE UNDERLYING SUBCLASSES.

Find the subclass before you compute anything else. Every summary statistic you calculate before you split is an average of two populations that do not belong in the same average.

04

HOW TO MEASURE SPREAD

STANDARD DEVIATION

Roughly the average distance of a data point from the mean. Small means the values cluster; large means they scatter.

The square root of the mean squared distance from the mean. Squaring is what stops the positive and negative distances from cancelling, and it is also why one extreme value moves an SD more than it moves a median.

COEFFICIENT OF VARIATION

The standard deviation divided by the mean, so spread can be compared across statistics measured on completely different scales.

CV = SD / mean. Unitless, so it survives a change of units. Meaningless when the mean sits near zero, and misleading on anything that can go negative.

? WHAT LEAGUE HAS THE MOST VARIATION IN SCORING?

THE RAW NUMBERS: SCORING PER GAME, LEAGUE BY LEAGUE

LEAGUE	STATISTIC	MEAN	SD
MLB	Runs scored	4.3	0.508
NFL	Points scored	22.2	5.49
NBA	Points scored	98.5	4.04
NHL	Goals scored	2.8	0.268

Rank these leagues by raw SD and decide whether that ranking makes sense, given how differently each league scores.

Severini (2020), Analytic Methods in Sports

05

SPREAD WITH NO NUMBERS TO AVERAGE

Standard deviation needs quantities. Play calls, pitch types, and formations are categories, and they need a different measure.

YOU CANNOT TAKE THE STANDARD DEVIATION OF “RUN, PASS, PASS, RUN.”

There is nothing to average. But a play-caller who runs on every short-yardage snap is obviously more predictable than one who splits it fifty-fifty, so the variation is real and it needs a number.

ENTROPY MEASURES HOW SPREAD OUT A SET OF SHARES IS

$H = -\Sigma\ p \cdot \ln(p)$ over every category

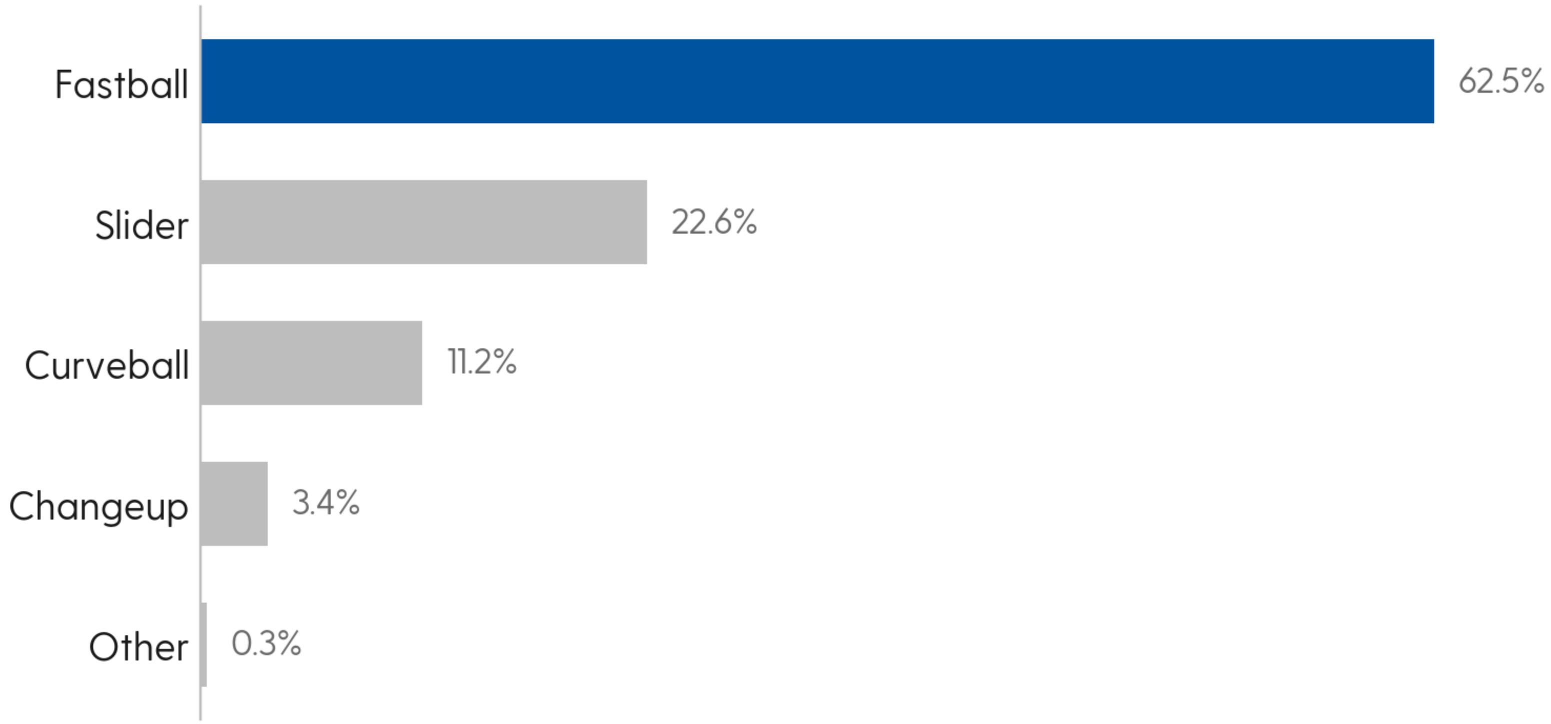
Max $H = \ln(n)$ the value when every share is equal

Standardized entropy $= H / \ln(n)$

In plain English: p is one category's share of the total (a run rate of 60% is p = 0.60), ln is the natural log (=LN() in Excel), Σ means add up one p · ln(p) term per category, and n is how many categories there are. A category with a share of zero contributes zero, not an error. Do not try to take the log of nothing.

RANGES 0 TO 1: 0 IS FULLY PREDICTABLE, 1 IS AN EVEN SPLIT.

KERSHAW THREW ONE PITCH 62.5% OF THE TIME



Two more pitches, the sinker and the cutter, are in the arsenal and were thrown zero times in this sample. Whether they count toward n is a decision you make and then state.

Published Statcast pitch-mix shares

ON THE ZERO-TO-ONE SCALE, WHERE DOES THIS PITCH MIX LAND?

06

THIS WEEK'S WORK

A hand-build Thursday, one case study with the agent, and a result that does not come out the way you would expect.

THURSDAY, IN CLASS: TWO HAND-BUILDS. THE DATA IS ALREADY IN YOUR REPO.

1. **NBA FG%: mean, SD, and coefficient of variation, in Excel.**
   Ten players, two per position, in weeks/week01/data/hand-build-nba-fg.csv. Sketch the shape while you are in there.

2. **NFL entropy: one team, on paper, from the formula.**
   One team's run/pass plays on third and fourth and short, in weeks/week01/data/hand-build-nfl-entropy.csv.

3. **Keep both.**
   They go in weeks/week01/ in your repo, and the agent has to match them before you accept anything it builds.

CASE STUDY: ENTROPY IN THE NFL

1. **Thursday, by hand: one team.**

   Standardized entropy for a single team's run/pass mix on third and fourth and short, from the formula, on paper. The plays are in your repo at weeks/week01/data/hand-build-nfl-entropy.csv. Keep it.

2. **Take-home, with the agent: all thirty-two.**

   Pull the play-by-play with the provided script, and have the agent explain to you how that script limits and caches its requests. Filter, compute each team's split, entropy, and conversion rate.

3. **Verify before you accept anything.**

   The agent has to reproduce your Thursday team and match your number. If it cannot, do not accept the league table.

4. **Ship the table and one chart.**

   Team table to outputs/ as .xlsx, plus a chart relating entropy to conversion rate. Then /audit, then /quiz-me, then /submit.

? COULD A TEAM'S ENTROPY COME OUT AT EXACTLY ZERO? WHAT WOULD THAT TAKE?

? DO THE LEAST PREDICTABLE TEAMS CONVERT SHORT YARDAGE MORE OFTEN?

WHAT GOES IN THE BRIEF.

Do you see a trend between standardized entropy and first down conversion percentage?

What other information would you want to look at to determine how important entropy is for success rate?

How would you go about assessing the optimal entropy for a given team?

How else would you analyze FG% to compare scoring efficiency across different positions?

Half a page, in BRIEF.md, due Monday at 11:59pm.

THE TAKE-HOME, STEP BY STEP

1 The question.

Do less predictable play-callers convert short yardage more often?

2 The data.

nflverse play-by-play, 2024 season. The pull script is already in your repo at weeks/week01/data/.

3 Open Claude Code in your repo.

Spec the job in your own words: the question, the target, the assumptions. (/teach is optional, if you want the week's concepts walked through first.)

THE TAKE-HOME, STEP BY STEP (CONTINUED)

4 Direct the agent.

Run the pull script, filter to third and fourth and short, compute each team's run/pass split, standardized entropy, and conversion rate.

5 Verify against Thursday.

Put your hand-build into weeks/week01/ in your repo: the Excel file, or a photo of the paper work. The agent must reproduce your team and match your number before you accept the league table.

6 Ship it.

Save the team table as .xlsx in outputs/, plus one chart. Write BRIEF.md in your own words. Run /audit and /quiz-me, then /submit, which checks and pushes. Canvas link by Monday 11:59pm.

BEFORE THURSDAY

Claude Pro, Claude Code, and Git installed. Your repo created from the template, with me added as collaborator, and MISSION.md filled in. That is your first commit.

Read the Week 1 quiz bank in your repo. It is not a secret, and the quiz comes out of it.

Bring your laptop with Excel, and paper for the entropy calculation. Finish the hand-build early and you start the case study in class instead of at home.